Bing-yue Wu

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Education

Arizona State University 2023 – 2026
Doctoral Studies in Electrical Engineering *Tempe, Arizona*
- Ph.D. candidacy achieved, 2026

National Taiwan University of Science and Technology (Taiwan Tech, NTUST) 2021 – 2023
M.S. in Electrical Engineering *Taipei, Taiwan*

National Taiwan University of Science and Technology (Taiwan Tech, NTUST) 2016 – 2020
B.S. with major in Electrical Engineering and minor in Computer Science and Information Engineering *Taipei, Taiwan*

Skills

Programming Languages: C, C++, Python, Tcl
EDA Tools: OpenROAD, OpenSTA, Innovus, Genus, ICC2, Design Compiler, HSpice, Virtuoso, Calibre
AI Tools: Claude Code CLI, Codex CLI, Cursor CLI
Language Ability: Mandarin (Native), English (Fluent)

Publications

B.-Y. Wu, C.-T. Ho, H. Yang, B. Khailany, and V. A. Chhabria, "**EvoDRC: A Self-Evolving Agentic Framework for Automated DRC Violation Repair**", *in Proc. ASP-DAC*, 2027.

V. A. Chhabria, A, Dey, A. B. Kahng, I. Savidis, P. Shrestha, and **B.-Y. Wu**, "**ResizerAgent: Agentic Strategy Selection for Timing Optimization in OpenROAD Resizer**", *in Proc. MLCAD*, 2026.

Z. Jiang, Q. Wu, **B.-Y. Wu**, and J. Zhang, "**CAPO: Certification-Guided Agentic Workflow for Physical Design Parameter Optimization**", *in Proc. GLSVLSI*, 2026.

B.-Y. Wu, C.-T. Ho, H. Yang, C.-C. Chang, A. B. Akkur, B. Khailany, and V. A. Chhabria, "**Special Research Session: Toward Agentic Solution for DRC Challenges in Digital VLSI Design**", *in Proc. DAC*, 2026. [GitHub]

B.-Y. Wu, A. Dey, A. Rovinski, and V. A. Chhabria, "**Focus Session: Large Language Models in Physical Design: From Data Generation to Intelligent Agents**", *in Proc. DATE*, 2026.

B.-Y. Wu and V. A. Chhabria, "**DALI-PD: Diffusion-based Synthetic Layout Heatmap Generation for ML in Physical Design**", *in Proc. ASP-DAC*, 2026. [GitHub]

V. A. Chhabria, A. Ghose, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Y. Liu, Z. Wang, and **B.-Y. Wu**, "**IEEE DATC RDF-2025: Enabling an EDA Research Ecosystem**", *in Proc. ICCAD*, 2025.

B.-Y. Wu, U. Sharma, A. Rovinski, and V. A. Chhabria, "**OpenROAD Agent: An Intelligent Self-Correcting Script Generator for OpenROAD**", *in Proc. ICLAD*, 2025. [GitHub]

V. A. Chhabria, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Z. Wang, **B.-Y. Wu**, and D. Yoon, "**Strengthening the Foundations of IC Physical Design and ML EDA Research**", *in Proc. ICCAD*, 2024.

V. A. Chhabria, **B.-Y. Wu**, U. Sharma, K. Kunal, A. Rovinski, and S. S. Sapatnekar, "**Generative Methods in EDA: Innovations in Dataset Generation and EDA Tool Assistants**", *in Proc. ICCAD*, 2024.

B.-Y. Wu, R. Liang, G. Pradipta, A. Agnesina, H. Ren, and V. A. Chhabria, "**2024 ICCAD CAD Contest Problem C: Scalable Logic Gate Sizing Using ML Techniques and GPU Acceleration**", *in Proc. ICCAD*, 2024. [GitHub]

U. Sharma*, **B.-Y. Wu***, S. R. D. Kankipati, V. A. Chhabria, and A. Rovinski, "**OpenROAD-Assistant: An Open-Source Large Language Model for Physical Design Tasks**", *in Proc. MLCAD*, 2024. [GitHub]

V. Gopalakrishnan, **B.-Y. Wu**, and V. A. Chhabria, "**ML-INSIGHT: Machine Learning for Inrush Current Prediction and Power Switch Network Improvement**", *in Proc. ISLPED*, 2024.

B.-Y. Wu, U. Sharma, S. R. D. Kankipati, A. Yadav, B. K. George, S. R. Guntupalli, A. Rovinski, and V. A. Chhabria, "**EDA Corpus: A Large Language Model Dataset for Enhanced Interaction with OpenROAD**", *in Proc. LAD*, 2024. **(Best Paper Nominated)** [GitHub]

V. A. Chhabria, W. Jiang, A. B. Kahng, R. Liang, H. Ren, S. S. Sapatnekar, and **B.-Y. Wu***, "**OpenROAD and CircuitOps: Infrastructure for ML EDA Research and Education**", *in Proc. VTS*, 2024. **(Primary Author)** [GitHub]

B.-Y. Wu, S.-Y. Fang, H.-W. Chang, and P. Wei, "**SpeedER: A Supervised Encoder-Decoder Driven Engine for Effective Resistance Estimation of Power Delivery Networks**", *in Proc. MLCAD*, 2022. **(Best Paper Award)**

Work Experience

Tenstorrent — Sept. 2026 – Present
Advanced Physical Design Intern — *Austin, Texas*

Arizona State University — Aug. 2023 – Aug. 2026
Graduate Research Assistant — *Tempe, Arizona*

- Developed a large-scale LLM dataset for physical design tool (OpenROAD) interaction, built a domain-specific LLM to generate OpenROAD scripts and answer OpenROAD-related queries, and created a self-correcting agent that iteratively refines scripts using real tool feedback.
- Proposed a diffusion-based generative framework to generate synthetic circuit layout heatmaps and open-sourced 23K+ synthetic samples to address data scarcity in ML-driven physical design research.
- Investigated LLM-based automated design rule violation (DRV) fixing and developed benchmarking infrastructure to evaluate LLM-driven DRC methodologies.
- Served as Topic Chair for ICCAD 2024 CAD Contest (Problem C) and contributed to OpenROAD/CircuitOps infrastructure to advance ML-EDA research and education.

Synopsys Inc. — Oct. 2021 – June 2022
Intern (Technical-Engineering) — *Taipei, Taiwan*

- Researched a novel Machine Learning-based (ML) solution estimating the effective resistance of Power Delivery Networks in advanced VLSI designs to speed up runtime and improve the accuracy of ML-driven IR analysis tools.
- Responsible for designing the data pipeline, the ML model architecture, and the entire effective resistance estimation workflow.

Academia Sinica — Jul. 2020 – Nov. 2020
Full-time Research Assistant at Computational Finance and Data Analytics Lab — *Taipei, Taiwan*

- Conducted research on a novel Transformer Encoder-based model with financial number category awareness, designing new pre-training and fine-tuning tasks to enhance its performance.
- Developed an Online Loan Application Recommender System for E.SUN Commercial Bank, achieving nearly 300% performance improvement. Responsibilities included workflow design, model development, and building Python APIs for industrial deployment.

Awards

Qualcomm Innovation Fellowship North America (QIF) — May 2026
MLCAD Student Travel Grant — Sept. 2024
Ferdinand A. Stanchi Fellowship — Aug. 2024
DAC Young Fellow Travel Grant — Jun. 2024
MLCAD Student Travel Grant — Sept. 2023
Fulton Fellows Fellowship — Aug. 2023
Best Paper Award at MLCAD 2022 — Sept. 2022

Professional Service

Reviewer Activities
ACM TODAES (2025×2), IEEE TCAD (2026×3)

Topic Chair of Problem C at 2024 ICCAD CAD Contest — Oct. 2024
IEEE CEDA — *Newark, New Jersey*

2024 ASP-DAC Tutorial Talk — Jan. 2024
ACM SIGDA — *Incheon, South Korea*